THE STALK OF THE ANNIHILATOR OF A MODULE OF FINITE TYPE IS THE ANNIHILATOR OF ITS STALK

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The inclusion between the stalk of the annihilator ideal and the annihilator of the stalk, valid without any finiteness hypothesis. The displayed inclusion referred to in the counted proof (source lines 4139 and 4119) is an earlier labelled equation of this chapter. Reduces the statement to the reverse inclusion, which is what the proof establishes.
- The injectivity of the canonical map from the stalk of the sheaf of homomorphisms into the homomorphisms of the stalks for a module of finite type. Invoked at the start of the counted proof (source lines 4131-4135) as a pointer to a lemma of this chapter. Converts the vanishing of a section on the stalk into its vanishing on a neighbourhood.

Source reference. Original equation equation-inclusion-of-annihilator-ideals: sources/stacks-modules/pinned-original-source.tex, source lines 4119–4119. Navigation pointer only: the counted text refers to the Stacks item equation-inclusion-of-annihilator-ideals, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-stalk-internal-hom: sources/stacks-modules/pinned-original-source.tex, source lines 3946–3946. Navigation pointer only: the counted text refers to the Stacks item lemma-stalk-internal-hom, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. If $\mathcal{F}$ is of finite type, then $(Ann_{\mathcal{O}_X}(\mathcal{F}))_x = Ann_{\mathcal{O}_{X,x}}(\mathcal{F}_x)$.*

Proof. By Lemma lemma-stalk-internal-hom the map

$$\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{F})_x \longrightarrow \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{F}_x, \mathcal{F}_x)$$

is injective. Thus any section f of $\mathcal{O}_X$ over an open neighbourhood U of x which acts as zero on $\mathcal{F}_x$ will act as zero on $\mathcal{F}|_V$ for some $U \supset V \ni x$ open. Hence the inclusion (equation-inclusion-of-annihilator-ideals) is an equality. □